Phonons and phonon symmetry

Jay Shah

1 Phonons and Phonon Symmetry

Phonons are quasiparticles that describe the collective vibrational excitations of atoms in a crystal. They are important for understanding heat transport, superconductivity, and phase transitions among other phenomena of condensed matter systems. The mathematical framework for phonons established using the Born-Oppenheimer (adiabatic) approximation is presented in Section 2 [1]. DFT is used to calculate the phonon modes of a crystal, and their frequencies can take real or imaginary values, with the latter indicating the structure is dynamically unstable. These phonon modes can be indexed by analyzing their symmetries.

Group theory is crucial to illustrate how phonon modes provide sample identification, and so a brief overview is provided. The symmetry operations of a crystal form a group [2]. Each symmetry operation can be represented by a transformation matrix which maps an initial set of coordinates onto another set after it has undergone the symmetry operation. Such a matrix is called a representative. Different basis functions, while having the same symmetry, may have different representatives. The complete set of representatives for a group is called its representation. Each representation spans its chosen basis. If all the representatives for a basis can be written in the same block diagonal form, such that each of the representatives can be written as the direct sum of two or more representatives of lower dimensionality, the representation has been reduced. When representations cannot be reduced anymore, they are called irreducible representations or irreps of the group. As the representation is reduced to form irreps, the original basis function has also been converted into a linear combination. This linear combination is called the symmetry adapted linear combination (SALC).

There are a limited number of symmetry species of a group. In other words, there are a limited set of ways that an arbitrary function can transform due to the symmetry operations of the group. Each of these symmetry species are irreps, and they are labelled according to their dimensionality and their behavior under certain symmetry operations. Each of the SALCs only transforms as one irrep. Also, one or more of the SALCs may transform as the same irrep. The 'Wonderful Orthogonality Theorem' (WOT) is used to identify the irreps spanned by a basis, and projection operators are used to identify SALCs that transform as each of the irreps. Details can be found in the text by Dresselhaus et al. [2]

Phonon modes are indexed in this manner, based on how they transform due to the symmetry operations of the space group. The difference is instead of using atoms as the basis, a $3N$ Cartesian basis is used, in which a set of x-, y- and z-axes are centered on each atom. Using the WOT and projection operators, which irreps are spanned by the Cartesian basis and which SALCs transform as the irreps, can be identified. The SALCs in this case are the atomic displacements associated with the phonon mode. Specifically, only information on which atoms are displaced and how is obtained. DFT is still required to compute the atomic distortions. SMODES, a program in the ISOTROPY Software Suite that can be run online or downloaded, gives the SALCs (also called the symmetry adapted modes) for all the phonon modes at the desired reciprocal lattice **k**-wavevectors

for a particular crystal [3]. DFT is required to obtain the frequency and displacements associated with the phonons.

2 Phonons

Using the Born-Oppenheimer (adiabatic) approximation a mathematical framework for phonons will be established [1]. The Hamiltonian of a periodic crystal as a function of atomic displacements from their equilibrium positions is:

$$\hat{H} = \Phi_0 + \frac{1}{2}\sum_i \sum_{N\nu} M_\nu \dot{U}_i(N\nu)^2 + \frac{1}{2}\sum_{ij}\sum_{MN\mu\nu} \Phi_{ij}\begin{pmatrix} M & N \\ \mu & \nu \end{pmatrix} U_i(M\mu)\, U_j(N\nu) + \frac{1}{3!}\sum_{ijk}\sum_{MNP\mu\nu\pi} \Phi_{ijk}\begin{pmatrix} M & N & P \\ \mu & \nu & \pi \end{pmatrix} U_i(M\mu)\, U_j(N\nu)\, U_k(P\pi) + \cdots \quad (2.1)$$

The first term Φ_0 is the ground-state electronic energy when there are no atomic displacements, and the second term is the kinetic energy due to nuclear motion. M, N and P are unit cell indices, μ, ν and π are atomic indices, and i, j and k are Cartesian directions. $U_i(M\mu)$ refers to the displacement of atom μ in unit cell M in direction i, $\Phi_{ij}\left(\begin{smallmatrix} M & N \\ \mu & \nu \end{smallmatrix}\right)$ is the second order force constant that couples displacements $U_i(M\mu)$ and $U_j(N\nu)$ to energy, and $\Phi_{ijk}\left(\begin{smallmatrix} M & N & P \\ \mu & \nu & \pi \end{smallmatrix}\right)$ is the third order coefficient. If the expansion is truncated to second order in atomic displacements, the potential energy of the crystal is given by:

$$F = \frac{1}{2}\sum_{ij}\sum_{MN\mu\nu} \Phi_{ij}\begin{pmatrix} M & N \\ \mu & \nu \end{pmatrix} U_i(M\mu)\, U_j(N\nu). \quad (2.2)$$

Force can be found by differentiating the potential energy with respect to displacement. Next, we assume solutions to the Hamiltonian are traveling waves with reciprocal lattice wavevector $\mathbf{k}$ and frequency ω such that:

$$U_i(M\mu) = \tilde{U}_i(M\mu)\, \exp\big(i\mathbf{k}\cdot\mathbf{r}(M\mu) - i\omega t\big) \quad (2.3)$$

The above expression can be differentiated twice with respect to time to find acceleration. And by using Newton's second law force = mass × acceleration, we obtain the following expression for force on atom $(N\nu)$ in the j^{th} direction:

$$\frac{\partial F}{\partial U_j(N\nu)} = \sum_{Mi\mu} \Phi_{ij}\begin{pmatrix} M & N \\ \mu & \nu \end{pmatrix} U_i(M\mu) = m_\nu\, \omega^2\, \tilde{U}_{j\nu}\, \exp\big(i\mathbf{k}\cdot\mathbf{r}(N\nu) - i\omega t\big) \quad (2.4)$$

where m_ν is the mass of atom $(N\nu)$. Next using Bloch's theorem, and defining a new index $L = M - N$, and performing some algebra, we get the expression:

$$\omega^2\, \tilde{e}_{j\nu} = \sum_{Li\mu} \frac{\Phi_{ij}\begin{pmatrix} L & 0 \\ \mu & \nu \end{pmatrix}}{\sqrt{m_\nu m_\mu}} \exp\big(i\mathbf{k}\cdot[\mathbf{r}(L\mu) - \mathbf{r}(0\nu)]\big)\, \tilde{e}_{i\mu} \quad (2.5)$$

where $\tilde{e}_{j\nu} = m_\nu^{1/2}\, \tilde{U}_{j\nu}$. We define the dynamical matrix $\mathbf{D}(\mathbf{k})$:

$$D(\mathbf{k})_{i\mu,j\nu} = \sum_L \frac{\Phi_{ij}\begin{pmatrix} L & 0 \\ \mu & \nu \end{pmatrix}}{\sqrt{m_\nu m_\mu}} \exp\big(i\mathbf{k}\cdot[\mathbf{r}(L\mu) - \mathbf{r}(0\nu)]\big). \quad (2.6)$$

The dynamical matrix satisfies the eigenvalue problem, where $\tilde{e}_{j\nu} = m_\nu^{1/2}\,\tilde{U}_{j\nu}$:

$$\omega^2\,\tilde{e}_{j\nu} = \sum_{i\mu} D(\mathbf{k})_{i\mu,j\nu}\,\tilde{e}_{i\mu} \tag{2.7}$$

$$\omega^2\,\tilde{\mathbf{e}} = \mathbf{D}(\mathbf{q})\,\tilde{\mathbf{e}} \tag{2.8}$$

where we have collapsed the atomic indices and Cartesian directions into a single compound index with a size 3 times the number of atoms in the unit cell, and $\mathbf{k}$ is replaced by $\mathbf{q}$ to distinguish between wavevectors for electronic states and phonons.

The eigenvalues of $\mathbf{D}(\mathbf{q})$ are the squared frequencies of each phonon mode, and their displacements are defined by $\tilde{e}$. Since the units of $\tilde{e}$ are length $\times$ mass$^{1/2}$, the true displacements require a division by the square root of the atomic mass. The eigenvalues can take positive or negative values, which means phonon frequencies can take real or imaginary values. Imaginary phonon modes indicate the system is dynamically unstable due to a displacement corresponding to that phonon mode. To solve for the phonon modes, the dynamical matrix needs to be calculated. To do so, a force constant matrix $\mathbf{F}(\mathbf{k})$ is defined:

$$F(\mathbf{k})_{i\mu,j\nu} \equiv \frac{\Phi_{ij}\begin{pmatrix} L & 0 \\ \mu & \nu \end{pmatrix}}{\sqrt{m_\nu m_\mu}} \exp\bigl(i\mathbf{k}\cdot[\mathbf{r}(L\mu) - \mathbf{r}(0\nu)]\bigr) \tag{2.9}$$

$$F(\mathbf{k})_{i\mu,j\nu} = m_\nu^{1/2}\,D(\mathbf{k})_{i\mu,j\nu}\,m_\mu^{1/2} \tag{2.10}$$

$$\mathbf{F}(\mathbf{q}) = \mathbf{M}^{1/2}\,\mathbf{D}(\mathbf{q})\,\mathbf{M}^{1/2}. \tag{2.11}$$

Phonon modes are computed by first calculating the dynamical matrix from the force constant matrix. The force constant matrix is calculated using the finite differences method:

$$\frac{\partial^2 E}{\partial U_j(N\nu)\,\partial U_i(M\mu)} = -\frac{\partial F}{\partial U_j(N\nu)} \tag{2.12}$$

where finite differences of forces equal the second order derivatives of total energy with respect to atomic position, calculated using DFT. There will be $3n$ phonon modes where n is the number of atoms in the unit cell. These phonon modes can be indexed by analyzing their symmetries.

3 References

[1] Dove, M.T., 1993. *Introduction to Lattice Dynamics* (No. 4). Cambridge University Press.

[2] Dresselhaus, M.S., Dresselhaus, G. and Jorio, A., 2007. *Group Theory: Application to the Physics of Condensed Matter*. Springer Science & Business Media.

[3] H. T. Stokes, D. M. Hatch, and B. J. Campbell, SMODES, ISOTROPY Software Suite, iso.byu.edu.